Northwest Geophysical Associates

GEOPHYSICAL REPORT

IN-LINE CSAMT SURVEY ON THE AFEK PROJECT

Golan Heights

IN-LINE CONTROLLED-SOURCE AUDIOMAGNETOTELLURIC SURVEY

Prepared for:

Genie Oil and Gas Company

November 14, 2013

NGA Job# 13060

TABLE OF CONTENTS

INTRODUCTION AND RESULTS SUMMARY 4

SURVEY METHODS 17

- IN-LINE AND BROADSIDE CSAMT 17
- MAGNETOTELLURICS 19

INSTRUMENTATION 19

DATA ACQUISITION 20

DATA PROCESSING 22

- SOFTWARE 22
- CSAMT PROCESSING FLOW 22

DATA QUALITY 22

DATA PRESENTATION 23

- PSEUDOSECTIONS 23
- SMOOTH-MODEL 1D INVERSION CROSS-SECTIONS 25
- DEPTH SLICE MAPS 28

DISCUSSION 28

- GEOLOGY 28
- GEOPHYSICAL INTERPRETATION 30

CONCLUSIONS 31

DATA DELIVERABLES 32

REFERENCES 32

APPENDICES*

Appendix A: Production Schedule

Appendix B: Transmitter and Receiver Station Coordinates

Appendix C: Resistivity Model Curves

Appendix D: Resistivity Model Cross Sections

**Separate document*

LIST OF FIGURES

Figure 1: Regional location of the Afek Project with polygon showing survey area. 6
Figure 2: Locations of CSAMT lines carried out in the Afek Project. 7
Figure 3: Dead Sea bitumen samples. 8
Figure 4: Forward modeling of the effects on frequency vs. resistivity curve shapes showing the difference between a 100 ohm-meter layer (top) compared to a 1000 ohm-meter layer (bottom) at varying depths, within in 10 ohm-meter homogeneous earth (provided by Genie). 9
Figure 5: Field data curves of frequency vs. resistivity for selected stations on Lines C2-4, E1S-4, K4-11, and K4-12 10
Figure 6: Contour map of the transition zone notch frequency (values in blue) end-on CSAMT survey lines. 12
Figure 7: Resistivity sections from cross-sections A-A' and B-B' (shown in Figure 2). 13
Figure 8: Resistivity plan view at a depth of 700 m below surface. 14
Figure 9: Resistivity plan view at a depth of 100 m below surface. 15
Figure 10: Elevation of 100 ohm-m contour in Afek Project area. 16
Figure 11: Comparison of sensitivity of broadside and end-on CSAMT arrays in detectionof a deep resistive layer. 18
Figure 12: Schematic of broadside CSAMT survey layout. 19
Figure 13: Sample apparent resistivity (a) and impedance phase (b) pseudosections 24
Figure 14: 1D inversion of CSAMT data on Line C2-3, station 500. Solid blue curve represents the modeled response to the data (+ symbols). Resistivity model is shown on the right side. 26
Figure 15: 1D resistivity cross-section of Line C2-3. 27
Figure 16: Synclinal structure of the Golan Heights. From Mor (1986). 28
Figure 17: Orthogonal measurements of resistivity along line M1 support a 1D interpretation. 30

CSAMT Survey on the Afek Project

INTRODUCTION AND RESULTS SUMMARY

Northwest Geophysical Associates (NGA) conducted a geophysical CSAMT (Controlled Source Electromagnetic) survey on the Afek Project with mobilization on September 06 and return on October 21, 2013.

The Afek Project is located in the southern Golan Heights (Figure 1). Sixty-two CSAMT survey lines, shown in Figure 2, were arranged according to transmitter and receiver access, the locations of which depended primarily on landowner permission. The goal of the survey was to use the CSAMT method to map subsurface high resistivity zones known to contain concentrations of kerogens, some of which have been thermally altered to bitumen (Figure 3), which can be further treated to produce liquid hydrocarbon products. Locally, the target zones are variously described as bituminous chalks (Rullkotter et al, 1984) or bituminous calcareous shales (Spiro and Rozenson, 1982), and are referred to as the Gareb, Upper Mishash, Mishash, and Menuha formations on lithologic logs provided by Genie.

NGA crew chiefs Tim Nordstrom and Josh Matthews supervised the field operations under Job #13060. Assistants included field technicians Paul Rodriguez, Jerry Huerstel, and Kyle Campbell. Program geophysicist Norm Carlson was responsible for project oversight from the NGA headquarters office, and provided 1D models of the processed data. Senior geophysicist Les Beard was responsible for CSAMT data processing. Geophysicist Anna Szidarovszky provided assistance in reporting.

The primary form of data collected in the southern Golan consisted of in-line CSAMT data with the transmitter several kilometers removed from the receivers. The CSAMT receiver lines were typically 800 m long, with electrodes spaced at 200 m intervals. Scalar in-line data were collected using combinations of 10 transmitters on 47 receiver lines. In addition, 15 lines of CSAMT data were collected with the transmitter arranged broadside to the receiver line. Finally, remote-referenced magnetotelluric (MT) data were collected at 5 locations.

This report covers data acquisition, instrumentation, data processing and preliminary interpretation of geophysical results. A summary of geophysical products provided with this report is given below:

- Survey location maps showing transmitter, line, and station locations
- Scalar-mode CSAMT plots

    - 1D inversion model resistivity profile sections
- Depth slices of 1D inverse models

A production schedule for the Afek Project is listed in Appendix A.

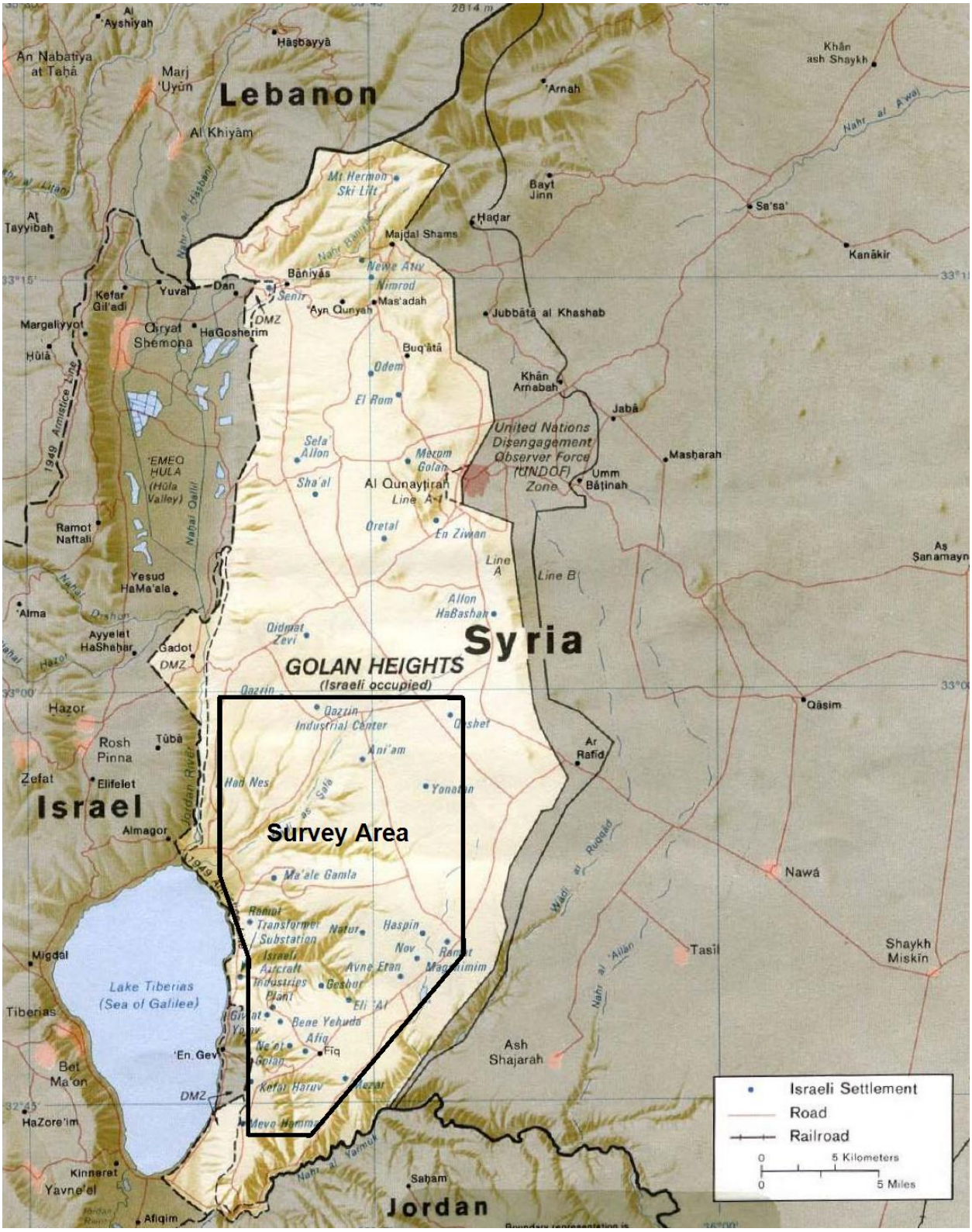


Figure 1: Regional location of the Afek Project with polygon showing survey area.

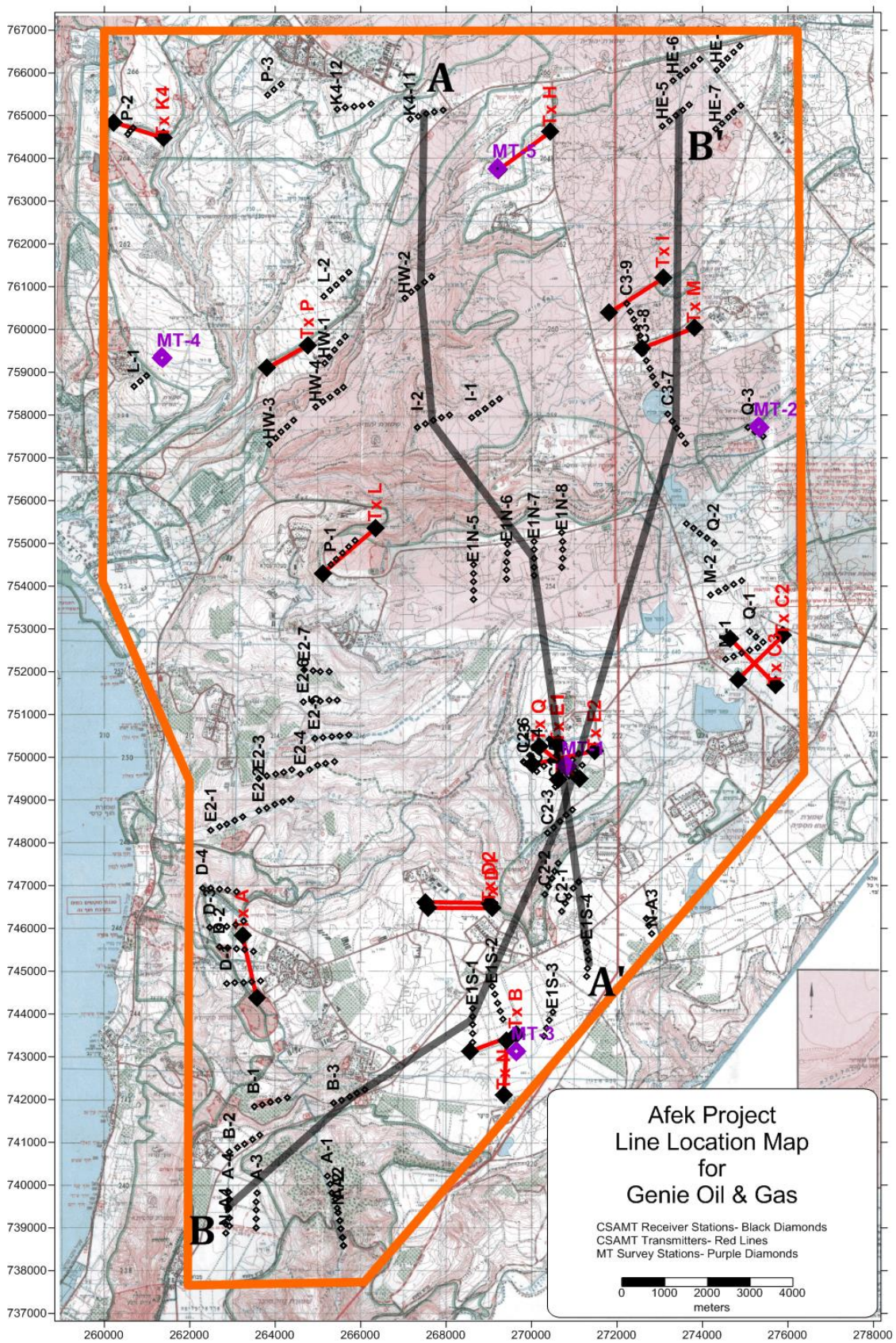
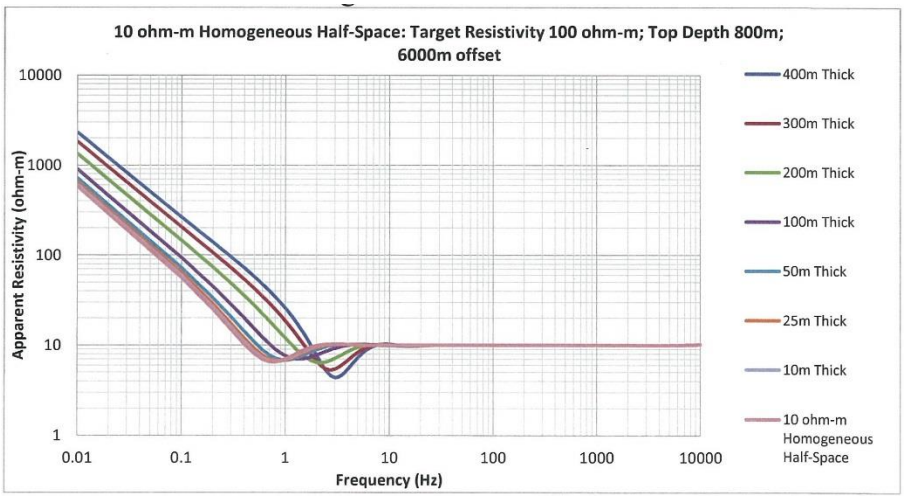


Figure 2: Locations of CSAMT lines carried out in the Afek Project.



Figure 3: Dead Sea bitumen samples.

Prior to the data acquisition, forward modeling by both Genie and NGA indicated that the in-line CSAMT geometry is significantly more sensitive to the targeted resistive layer(s) than broadside CSAMT or LoTEM methods. This increased sensitivity is evident as changes in the transition zone from "far-field" frequencies (high frequencies) to "near-field", low frequencies; this part of the frequency-versus-resistivity curve is called the transition zone notch. Figure 4 shows forward modeling results (Vinegar, 2013) for varying thicknesses of a 100 ohm-meter layer (top) as compared to a more resistive, 1000 ohm-meter layer (bottom). The more resistive layer produces an increasingly sharp notch near 10 Hz in the curve shape as the resistive layer is increased in thickness.



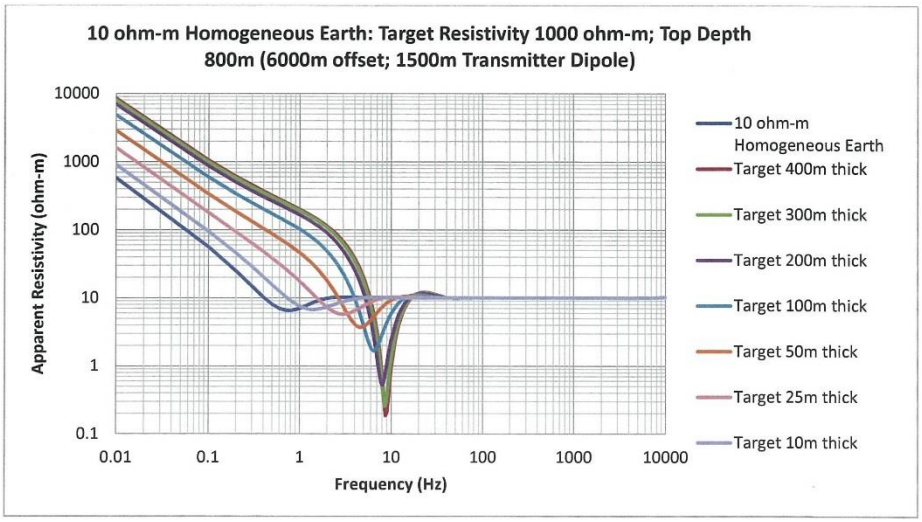


Figure 4: Forward modeling of the effects on frequency vs. resistivity curve shapes showing the difference between a 100 ohm-meter layer (top) compared to a 1000 ohm-meter layer (bottom) at varying depths, within in 10 ohm-meter homogeneous earth (provided by Genie).

Field data showed good correlation with the modeling results, with sharp transition zone notches in the south-central part of the survey area, as compared to smoother, lower frequency notches in the northwestern part of the survey area. Figure 5 below shows a comparison of field data curves from Lines C2-4 and ES1-4 (south-central survey area) versus Lines K4-11 and K4-12 in the northwest. The transition zone notch shifts from 24 Hz down to 1 Hz from the south-central lines to the northwestern lines.

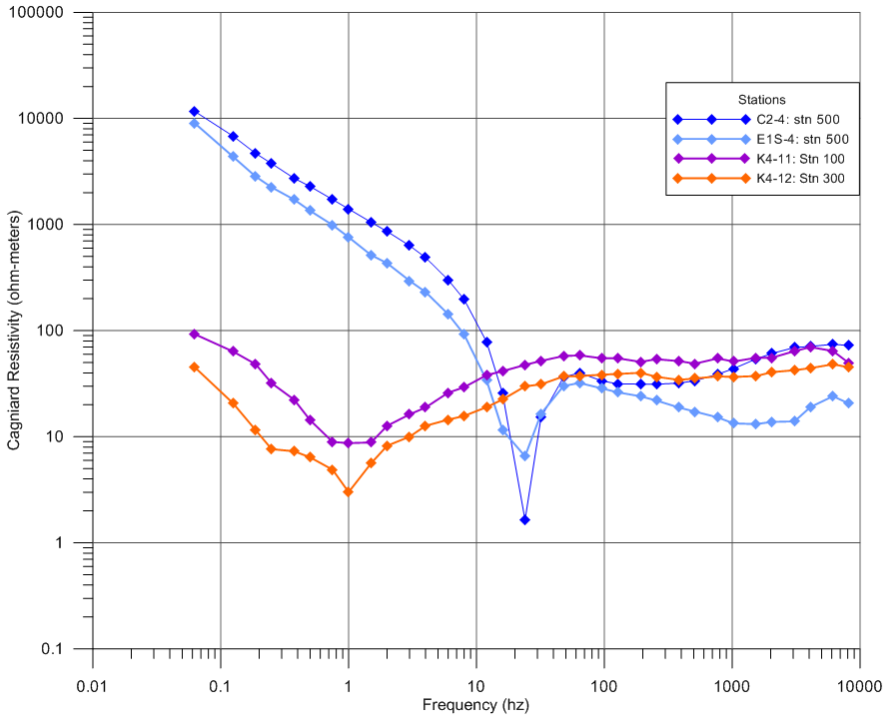


Figure 5: Field data curves of frequency vs. resistivity for selected stations on Lines C2-4, E1S-4, K4-11, and K4-12.

Plotted in map view, the behavior of the transition zone notch across the survey area provides a very coherent picture (Figure 6). High values, indicating the presence of a resistive unit, are seen in the south-central and eastern part of the survey area, while the lowest frequency notch zones are seen in the northwest, where seismic data suggests that the target resistor is very deep, if present. Low frequency values are also seen in the southwest corner of the survey area. It is interesting to note that the trends evident in this map view do not appear to be simply correlatable with surface or topographic features, or with conditions such as proximity to the Kinneret.

In addition to examination of the frequency vs. resistivity curves, inversion modeling of the data was also done to calculate resistivity vs. depth curves (Appendix C) and to produce resistivity cross sections (Appendix D) for each of the survey lines. We have chosen two sample sections, A-A' and B-B' (shown in Figure 2) across the survey area and have plotted resistivity cross sections for the lines in these sample sections. Both of these sections, shown in Figure 7, illustrate similar changes in the resistivity of the target unit–the Gareb-Menuha interval–from south to north. The target interval gets generally less resistive towards the north, possibly indicative of less thermal maturation of this zone toward the north.

Constant depth plan view resistivity slices are shown in Figures 8 and 9 for depths of 700 m and 100 m respectively. The 700 m depth slice cuts through much of the target interval and shows higher resistivities in the east than in the west. This suggests that either the target zone in the eastern part of the survey area is more thermally mature than in the west, or that it is substantially deeper in the west. In the northwest, the 700 m depth section is less than 10 ohm-m. This is consistent with the transition zone changes in Figure 6 and the cross-sections of Figure 7. The 100 m resistivity depth slice in Figure 9 is interesting in that it shows a high resistivity bullseye in the southeast part of the survey area. This may indicate a substantial volume of shallow, more easily accessible rock which has been thermally altered.

Shown in Figure 10 is a plan view plot showing the depth to the 100 ohm-m contour. It shows the 100 ohm-m resistivity contour to be substantially shallower in the east than in the west. In particular, the northwest part of the survey area shows a very deep 100 ohm-m contour, 1000 m below sea level or more. Inferences from this figure are consistent with those of previous figures: the target zone appears more thermally mature in the east, and probably shallower as well.

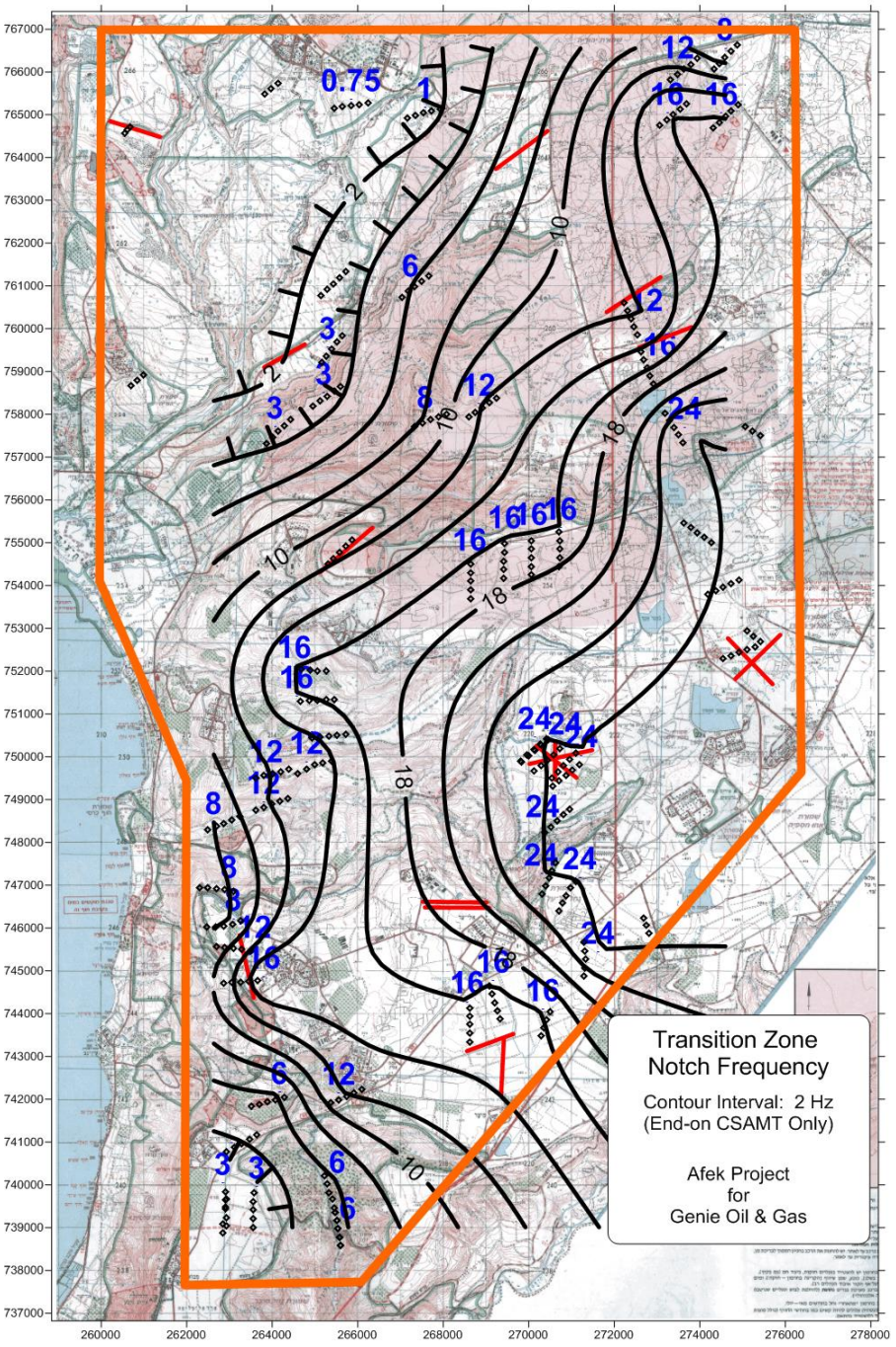


Figure 6: Contour map of the transition zone notch frequency (values in blue) end-on CSAMT survey lines.

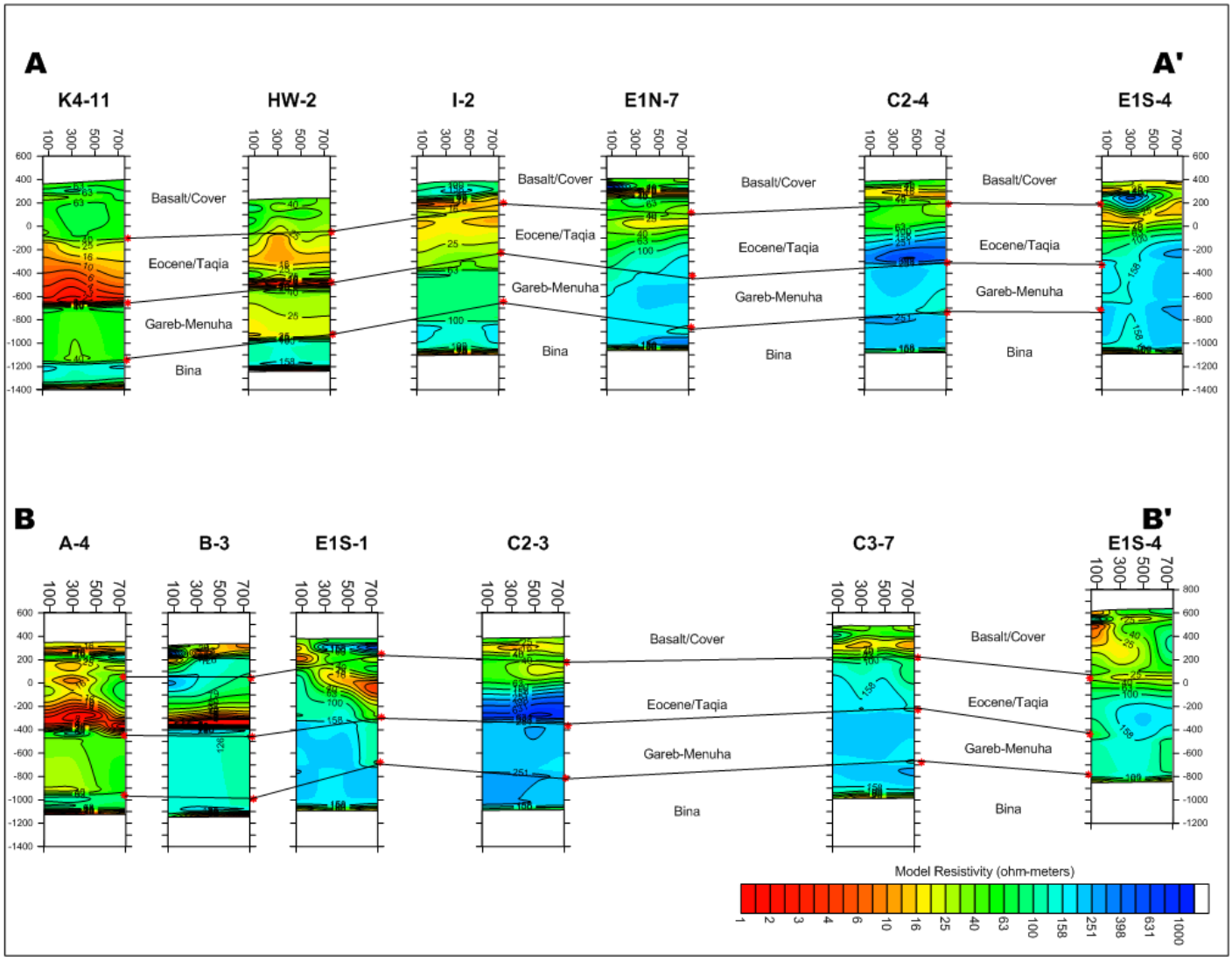


Figure 7: Resistivity sections from cross-sections A-A' and B-B' (shown in Figure 2).

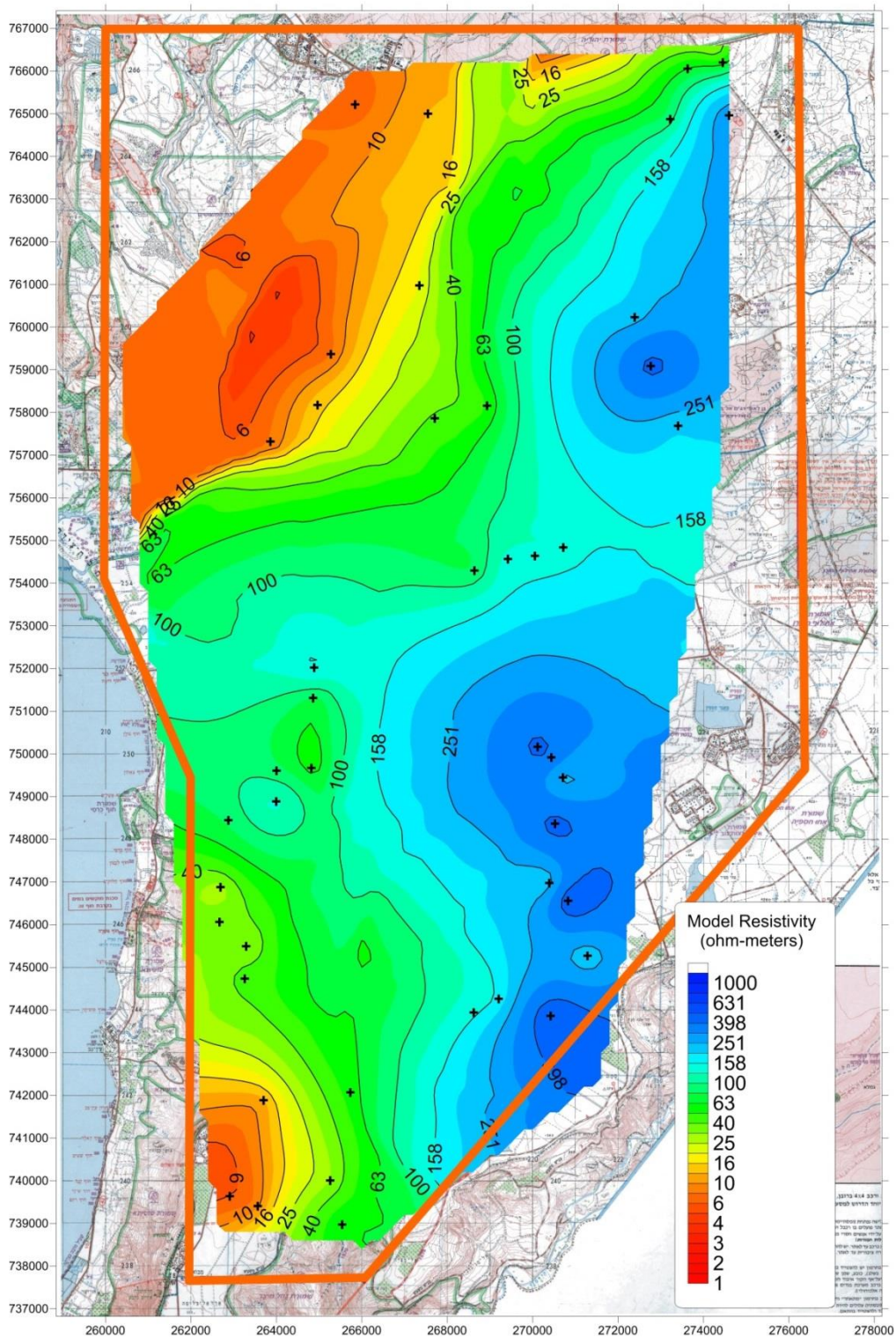


Figure 8: Resistivity plan view at a depth of 700 m below surface.

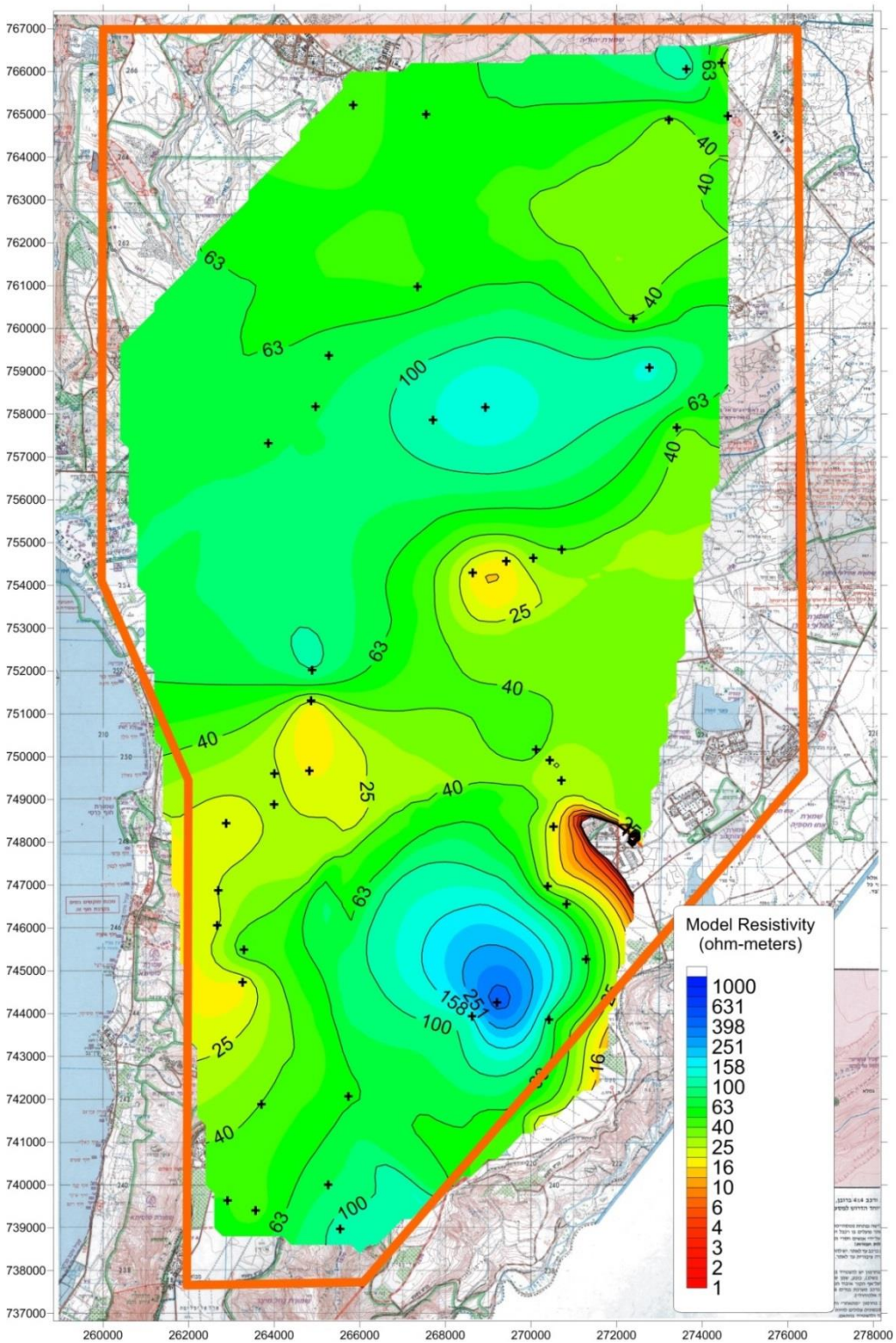


Figure 9: Resistivity plan view at a depth of 100 m below surface.

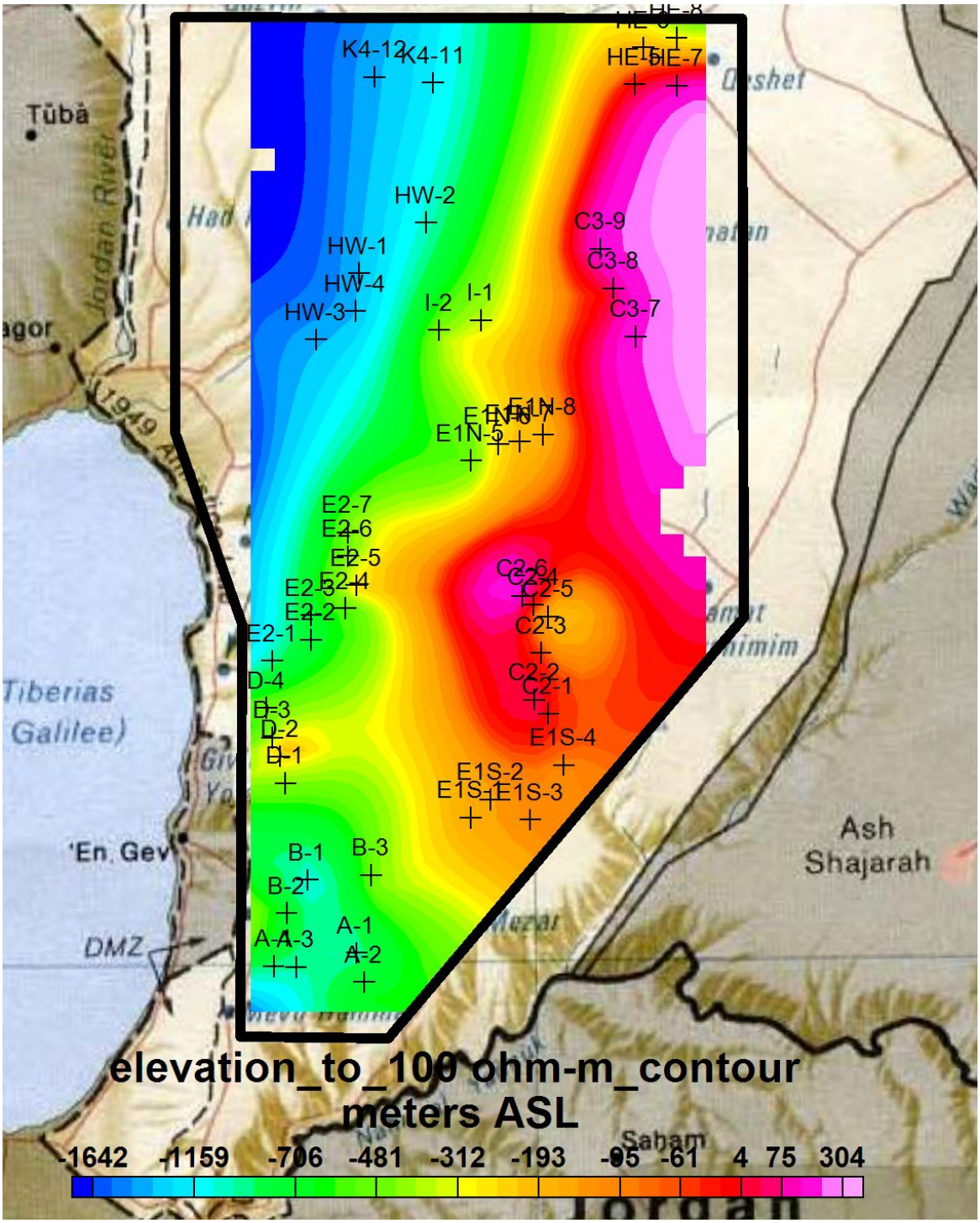


Figure 10: Elevation of 100 ohm-m contour in Afek Project area.

SURVEY METHODS

In-line and Broadside CSAMT

In-line controlled source audiomagnetotelluric (CSAMT) data were collected along 47 lines in the southern Golan survey area, shown in Figure 2. In addition, 15 lines of broadside CSAMT were collected. For both in-line and broadside configurations, the measured quantities are the magnitude and phase of the electric and magnetic fields along an array of grounded receiver dipoles. Most receiver lines were 800 m long with 200 m electrode spacing, though a few were shortened because of permitting limitations or culture. The transmitter length was 1500 m, where permitting allowed. The receiver measures the magnitude of the received signal and the absolute phase difference between the received signal and the transmitted signal, which is known via synchronization of the transmitter and receiver.

The Cagniard (apparent) resistivity is calculated from the magnitude of the fields by the relation $\rho_a = \frac{1}{5f}\left|\frac{E}{H}\right|^2_{\perp}$ and impedance phase is the difference of the absolute phases of the E and H fields and is given by $\phi = \phi_E - \phi_H$. In these equations ρ_a is the Caginard resistivity in ohm-meters, E is the electric field in mV/km, H is the magnetic field in nanoteslas (nT), Φ is the phase in milliradians, and f is the frequency in hertz.

A geophysical inversion algorithm can then be used to compute an estimated earth resistivity model from the frequency-based data. One advantage of the CSAMT method is that it provides much higher resolution, both laterally and at depth, than direct current electrical resistivity/IP methods, although CSAMT provides no induced polarization information.

A schematic setup diagram for broadside CSAMT field measurements is shown in Figure 12. CSAMT may be collected in either scalar or vector mode. Scalar mode measures only the along-line electric field, Ex, and the across-line magnetic field, Hy. In vector mode CSAMT, both components of the electric field and magnetic field are measured: the along- line components, Ex and Hx, and across-line components, Ey and Hy. By design, the transmitter is oriented so that electric current is directed parallel to survey lines and perpendicular to the assumed geologic strike. The bulk of the information for this type of survey is in the Ex and Hy components (scalar CSAMT), if the geology is generally 1D or 2D and the survey layout is optimized. In areas of very complex geology, measuring the complimentary components (vector mode) may allow better resolution of 3D features; however, the increase in survey cost can be

significant. Scalar CSAMT data were collected on the in-line CSAMT lines, and the broadside lines were collected with a portion of the data orthogonal to the main line.

Although the broadside CSAMT setup is more commonly used than the in-line, for this project we chose the in-line because it showed greater sensitivity to a deep resistive target. Figure 11 shows a four layer geoelectrical model in which a 1000 ohm-m layer is 720 m below more conductive units. The two curves show the frequency sensitivity of the two array types to the deep resistive layer. The in-line (or end-on) configuration has a clear advantage in sensitivity over the broadside array for a deep resistive target, as is the case with the Afek Project.

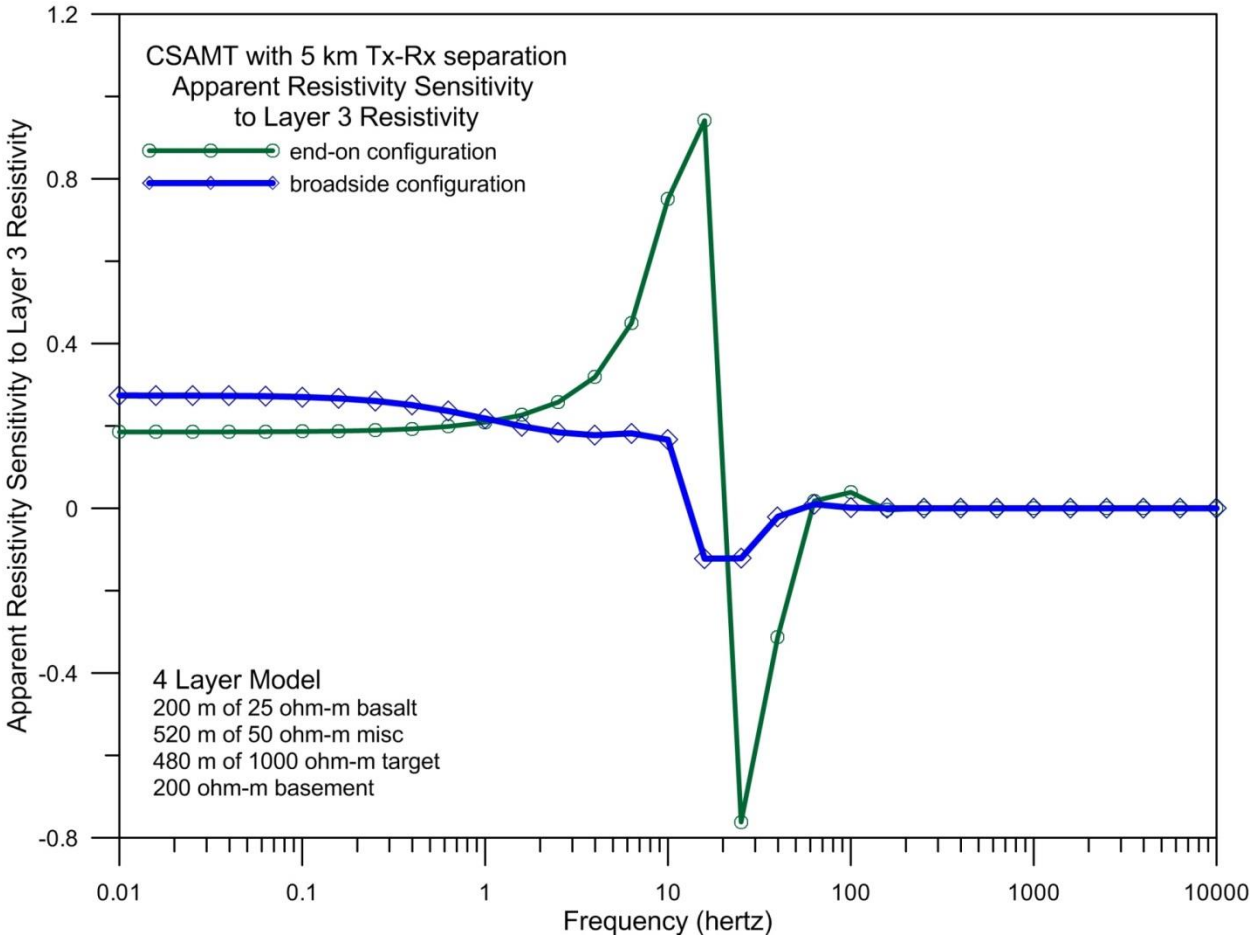


Figure 11: Comparison of sensitivity of broadside and end-on CSAMT arrays in detectionof a deep resistive layer.

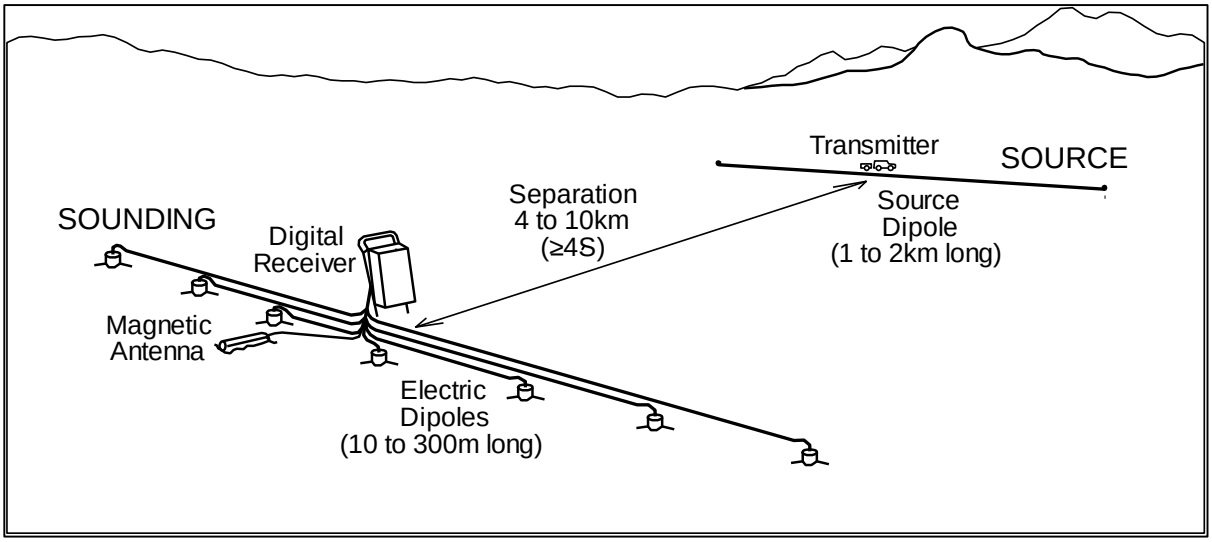


Figure 12: Schematic of broadside CSAMT survey layout.

Magnetotellurics

At the client's request, NGA collected magnetotelluric data at five different station locations, labeled MT-1 through MT-5 in Figure 2. A remote-reference station was used, but the data from these stations nevertheless proved to be very noisy. The MT data are not further discussed in this report, but will be addressed in a future report.

INSTRUMENTATION

A Zonge 8-channel GDP-32^{II} multi-purpose geophysical receiver was used for this survey. The GDP-32^{II} instrument is a backpack-portable, 16 bit, microprocessor-controlled receiver. The electric-field signals were measured using non-polarizing ceramic porous-pot electrodes connected to the receiver with insulated 14-gauge wire. Magnetic field data were acquired with two Zonge ANT/6 magnetic field sensors.

The signal source for this project was a Zonge 30 kilowatt GGT-30 transmitter. Transmitter power was provided by a Zonge ZMG-30 gasoline-powered motor-generator. The transmitter was controlled by the XMT-32 transmitter controller. Data collection requires time and frequency synchronization between the GDP-32^{II} and GGT transmitter. This synchronization process establishes a common time base between the transmitted and received

electrical signals. Zonge equipment specifications may be found on the report archive disk. The following are the serial numbers of the primary instruments used on this project:

• Receiver:	GDP 32^{II}	SN# 3220
		SN# 32284
		SN# 32287T
		SN# 3282T
• Transmitter:	GGT-30	SN# 2049
• Generator Set:	ZMG-30D	SN# 008
• Transmitter controller	XMT-32G	SN# G21
• Magnetic Antenna:	ANT/6	SN# 2526
		SN# 2516
		SN# 2486
		SN# 2126
		SN# 586

DATA ACQUISITION

All GPS coordinates and maps are provided in the New Israeli Datum (NID), Transverse Mercator projection, meters. Station positioning was achieved using a hand-held GPS receiver (Garmin Legend HCx). Planned station coordinates were generated by NGA along the proposed survey lines. These coordinates were loaded into the GPS receiver and the crew navigated and staked the actual survey stations prior to data acquisition. Where necessary, the stations were offset around obstructions or to accommodate permitting limitations.

In-line scalar mode CSAMT were acquired using 200 m station spacing along 47 lines at Golan. Station coordinates for all lines are listed in Appendix B.

Current electrodes at the transmitter site were constructed using steel stakes. Each transmitter electrode consisted of 20-30 stakes, connected by 14 gauge insulated wire and pounded approximately one-half meter into the earth and soaked with salt water to improve contact resistance.

Electric field measurements were made at the receiver site along spreads of potential electrodes consisting of grounded, non-polarizable porous pots filled with copper sulfate solution. The pots were partially buried in small pits and water was added to reduce contact resistance with the ground.

Data were typically collected using 4 channels per setup, or spread. The measurement spread usually consisted of 4 Ex dipole and 1 Hy magnetic field measurement. In this configuration, Ex is the electric field measurement made in the along-line x-direction (parallel to

the transmitter dipole), and Hy is the magnetic field measurement in the across-line or y-direction.

CSAMT frequencies from 8192 down to 0.125 or 0.0625Hz were acquired in the field. The entire suite of frequencies was used for 1D resistivity modeling (8192 – 0.0625 Hz).

DATA PROCESSING

Software

All raw, averaged and inversion model data and images are included on the Report Archive Disk. Zonge software was used for processing and modeling the Golan CSAMT survey data using the following programs:

- Zonge CSAVGW (v1.10u; for visualizing, averaging and editing raw data)
- Zonge ASTATIC (v3.60p; for visualizing and corrections of averaged data)
- Zonge SCS2D (v3.40h; for 1D inversion modeling)
- Zonge MODSECT (v4.42b; for plotting 1D model sections)
- Zonge SCSPLOT (v3.21c; for plotting model sections and pseudosections)
- Golden Software Surfer (v7.5p; for final database, plotting, visualization models)

CSAMT Processing Flow

A brief outline of data processing and modeling steps is given below:

- Prepared *.mde (Survey configuration and processing control file) and *.stn (station coordinate) files.
- Processed raw data files (.raw) using the *CSAVGW* program. CSAVGW is a CSAMT/NSAMT data averaging and quality-control utility program, displaying the data in pseudosection, sounding curve or data point graphs to allow skipping of outlier data. Data are saved as averaged data in *.avg files.
- Process averaged data with the *ASTATIC* algorithm using ASTATIC. Individual sounding curves are viewed and final removal of individual measurements that are considered to be affected by noise is made.
- Performed 1D inversion modeling using *SCS2D* smooth model inversion programs. Ran *SCS2D* using *Ex.avg files to get 1D model resistivity cross-section (depth profile) models.
- Model results (*.mtm, 1D models) were imported into Golden Software Surfer for plotting.

DATA QUALITY

Data quality is monitored in the field by the operator. Real-time standard-error values are displayed during acquisition. In addition, multiple measurements at a range of frequencies are displayed graphically as resistivity versus frequency curves with error bars showing the data scatter. This allows a visual evaluation of the data quality and remedial action to be taken if

necessary. Data are further evaluated for quality during the standard data reduction process whereby spurious data are removed if necessary.

Standard NGA field procedure requires that the receiver operator make multiple measurements (stacks) of each data point while monitoring receiver displayed real-time standard-error values. Provided sufficient stacks are collected in the field, the averaged sounding curves may be manually edited by removing outliers and noise that remain in the data after the automated, robust averaging technique is applied. The survey data were collected using 8-16 stacks at low frequencies resulting in data quality sufficient to address the goals of the Afek Project.

One important influence on electrical geophysical data quality is the electromagnetic noise generated by human activity (referred to as culture). Typical cultural-related noise sources include RF and electrical power line transmissions, cathodic protection on pipelines, metal fences, and buried or surface pipelines in the area. Culture due to infrastructure were fairly significant on this project.

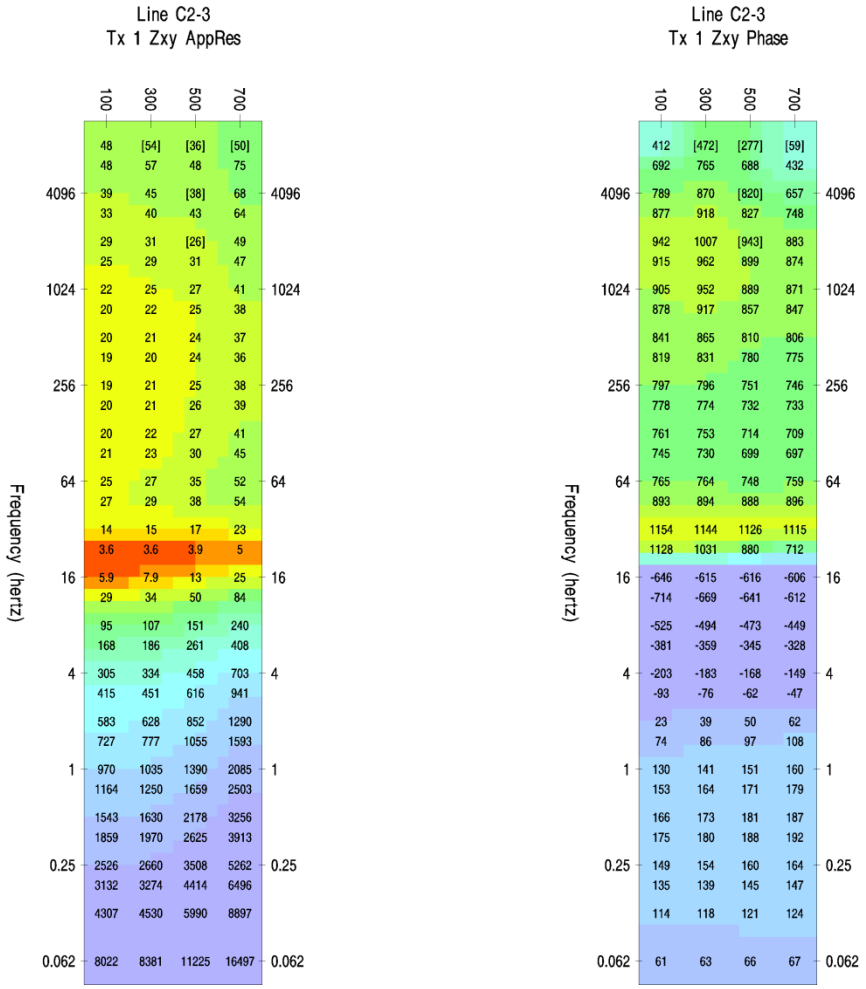
Raw measurements occasionally exhibited moderately high scatter, particularly the phase values, primarily caused by culture, including power lines. The modeled resistivities are not significantly affected by noise due to robust noise handling in the averaging and smooth-model inversion algorithms.

In general, the data processing, analysis, and model results indicate very good survey results at Golan.

DATA PRESENTATION

Pseudosections

Pseudosections are plots of station measurements versus frequency; in general the lower frequencies correspond to deeper depth of investigation. The pseudosections are presented as contoured plots of Cagniard Resistivity and Impedance Phase. In all pseudosections, the brackets [ ] indicate individual measurements affected by coherent noise and skipped in the processing flow. Figure 13 show the pseudosections for Line C2-3.



(a) (b)

Figure 13: Sample apparent resistivity (a) and impedance phase (b) pseudosections taken from Line C2-3.

Smooth-model 1D Inversion Cross-sections

The CSAMT survey data have been modeled using a 1D inversion algorithm to generate model cross-sections and plan view maps of the estimated subsurface resistivity. The resistivity inversion cross sections are computed using Zonge's SCS2D algorithm (MacInnes and Raymond, 2006), that fits the resistivity data to a one-dimensional smoothly varying, multi-layered, geoelectric model. Smooth-model inversion is a robust method for converting CSAMT data to resistivity model cross-sections. The forward calculations are based on a horizontally layered earth algorithm by Anderson (1974). The forward model is limited to 1D models, but the solution is valid over a full range of frequencies, encompassing the near-, transition-, and far-fields. It is important to note that mathematical inversion, by its nature, presents statistically optimized yet non-unique solutions. Geologic knowledge and constraints must always be incorporated when interpreting inverse models.

SCS2D inverts observed apparent resistivity and impedance phase data from a line of frequency-based CSAMT or natural source MT soundings to determine resistivities in a model earth cross-section. Either TM mode data, defined in this case as line parallel E-field, or TE-mode data, line perpendicular E-field, may be inverted.

During the inversion, model resistivities are adjusted iteratively until calculated apparent resistivity and impedance phases are as close as possible to observed data, consistent with model constraints. Model constraints include background-model constraints, which restrict the difference between the inversion model and a background model section, which represents known geology, and smoothness constraints, which limit resistivity variation from pixel to pixel. To calculate apparent resistivity and impedance phase for a given model section, SCS2D uses a layered earth algorithm to calculate far-field CSAMT or natural-source MT data (Anderson, 1974). Either TM- or TE-mode data can be calculated for scalar, vector or tensor survey configurations for frequencies ranging from less than 0.01 Hz to 10 kHz. For this project, a TM-mode inversion of array data was performed on both the in-line and broadside CSAMT lines.

The 1D inversion makes the assumption that geology is a uniformly layered, 1D earth at each station; individual stations are inverted independently. While the 1D inversion is robust and preserves detail, often illuminating sharp vertical features better than 2D models, 1D-modeled results can be distorted when data are strongly controlled by 2D and 3D geology.

For interpretive purposes, inversion resistivity models are presented in this report as both color-contoured cross sections and as color-strip, depth slice maps. In the model sections, cool colors (blues, greens) represent high resistivity and warm colors (orange, red, pinks) represent

low resistivities. Figure 14 shows an example of a 1D model for Line C2-3, station 500. The left panel shows the fit of the modeled response to the data, and the right hand panel shows the resistivity as a function of depth below surface. After all the stations are modeled, the solutions are adjusted to topography and collectively gridded to produce a section such as the one for Line C2-3, shown in Figure 15.

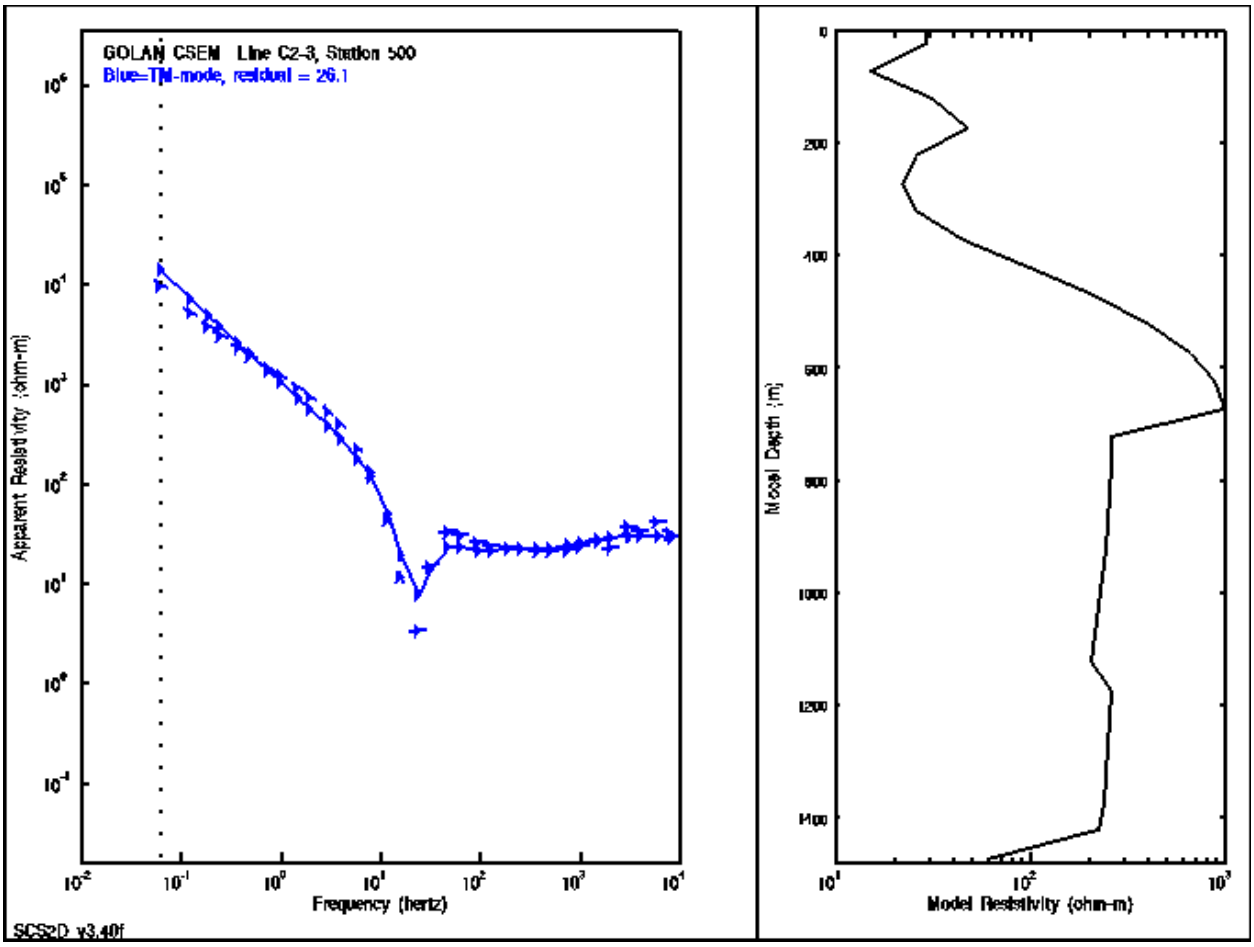


Figure 14: 1D inversion of CSAMT data on Line C2-3, station 500. Solid blue curve represents the modeled response to the data (+ symbols). Resistivity model is shown on the right side.

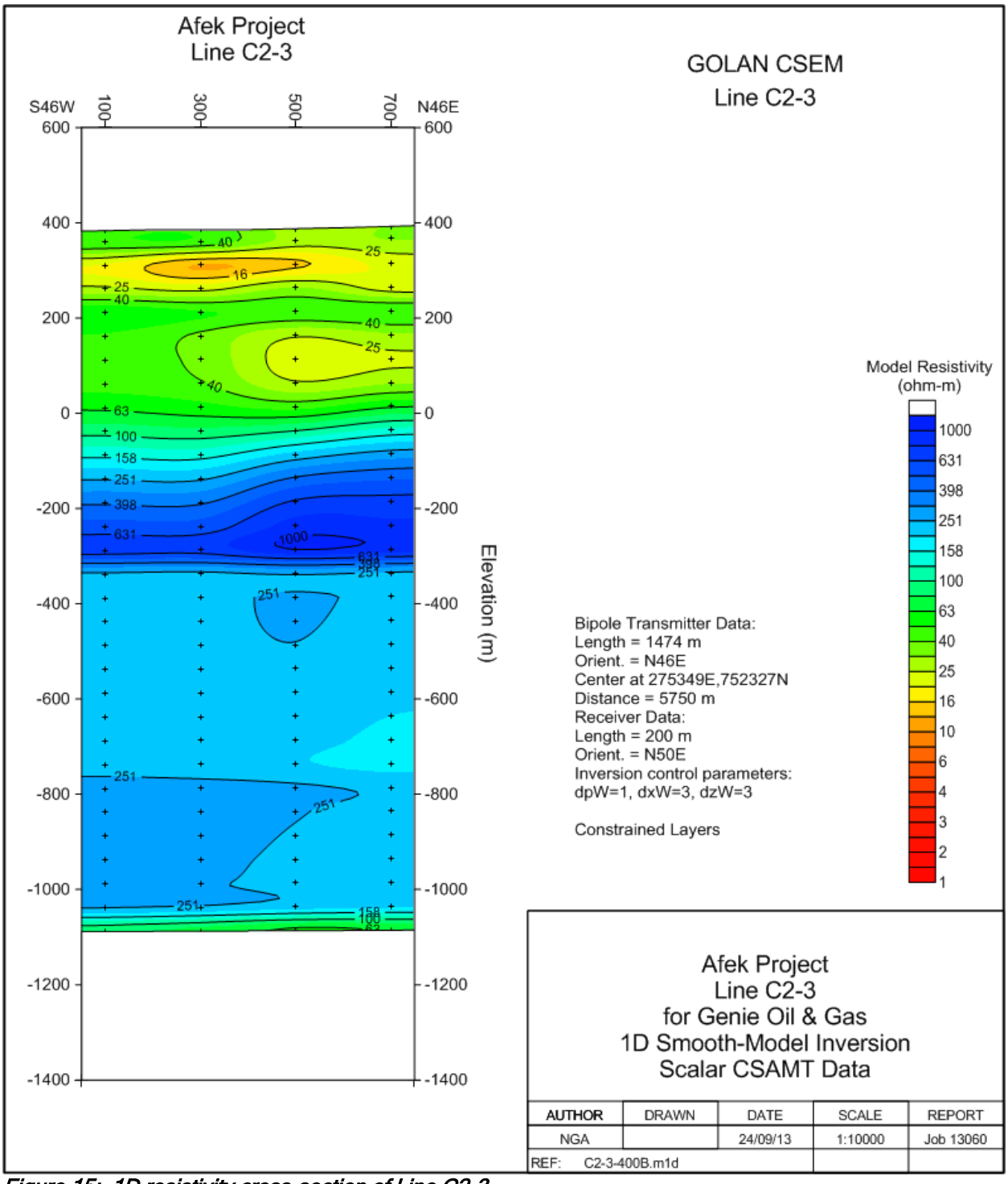


Figure 15: 1D resistivity cross-section of Line C2-3.

Depth Slice Maps

Plan-view depth slice maps were created from the 1D smooth-model inversions. The irregular distribution of the survey lines for this project made gridding problematic. However, the plan-view depth slice maps appear consistent with other types of maps and with what is known from previous investigations. Depth slice maps for topography-draped depths of 100 m and 700 m are shown in Figures 8 and 9 above.

DISCUSSION

Geology

The upper rock units of the Golan Heights proper are mostly basalts and other types of volcanic rock, forming a plateau that drops off to the west, to the Jordan River and Lake Kinneret, and to the south, to the Yarmouk River. The plateau is crossed by a number of seasonal streams which run through valleys, and flow west into the Jordan or Lake Kinneret.

According to Meiler (2011), the Golan Heights comprise a broad syncline with a surface layer of basalt flows overlying older sedimentary units, as illustrated in Figure 16, extracted from Mor (1986). On the scale of our survey lines, e.g. 800 m, the synclinal structure effectively appears as horizontal layers, or layers with minor dip. Results of seismic surveys presented by Meiler (2011) support the layered model, with some faults appearing in the sections.

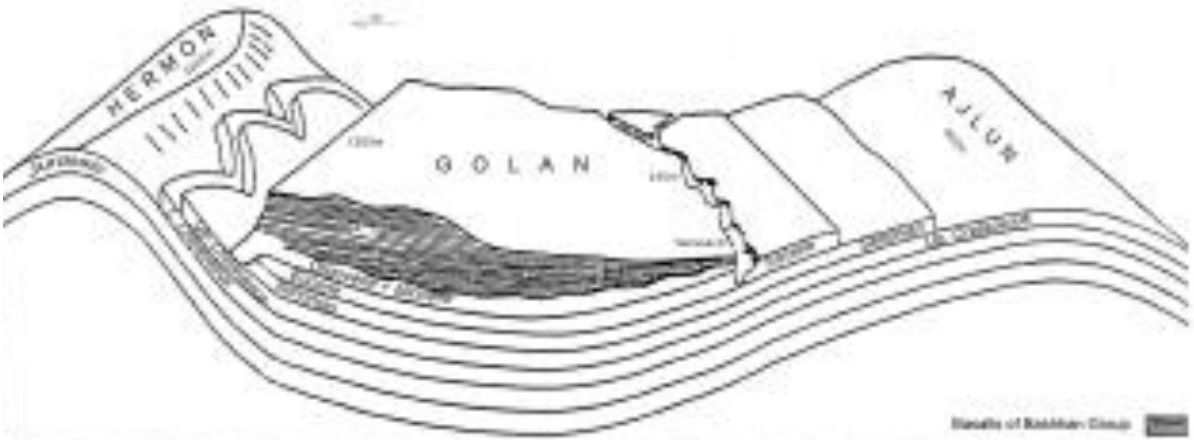


Figure 16: Synclinal structure of the Golan Heights. From Mor (1986).

The generalized geology of the survey area consists of surface basaltic units, produced from extrusive lava flows from the formation of the Red Sea rift graben, and part of the extensive Jebel al Arab basalt field. The basaltic unit has variable thickness, and in the survey area can range from a few tens to more than 500 m in thickness. The basalt unit unconformably overlays a series of limestone, siltstone, sandstone, chalk, and marl units that can range in total thickness from a few hundred meters to more than a kilometer. These units overlay the target units–the Gareb, Upper Mishash, Mishash, and Menuha formations. The total thickness of the target formations is usually a few hundred meters. The target units also may vary considerably in the amount of economically viable hydrocarbons they contain in the form of thermally converted kerogens.

Table 1 shows the names, lithologies, and resistivities of the vertical sequence of formations in the survey area.

Measured resistivities from the CSAMT survey range from low to moderately high, under 10 ohm-m to more than 1000 ohm-m. Background resistivity values are on the order of 50 to 150 ohm-meters in this survey area.

Table 1: Lithologies in the Afek survey area and their resistivities

Unit name	Lithology	Resistivity (ohm-m)
Holocene	alluvium	250
Cover Basalt	basalt	250
Gesher	limestone	15
Hordos	siltstone/sandstone	15
Fiq	chalk	15
Eocene	chalk	15
Taqia	marl	15
Gareb	chalk	4
Upper Mishash	limestone + shale	10
Upper Mishash Base	shale+limestone	10
Mishash	shale + limestone	35/1000
Menuha	chalk	200/1000
Bina	limestone	1000

Geophysical Interpretation

In general, the lengths of the in-line CSAMT receiver lines were short enough that the earth in the vicinity of the lines was effectively 1D. This enabled us to use Zonge's 1D CSAMT inversion software, which is valid for near-, transition-, and far-fields. Figure 17 shows data collected using electrically orthogonal sensors, allowing us to compute apparent resistivities ρ_{xy} and ρ_{yx}. Where these were approximately the same, the 1D approximation could be assumed valid. Although locally the earth could be approximated 1D, over larger distances it changed. We were able to map these changes by connecting several 1D receiver lines in a kind of fence diagram, as shown in Figure 7.

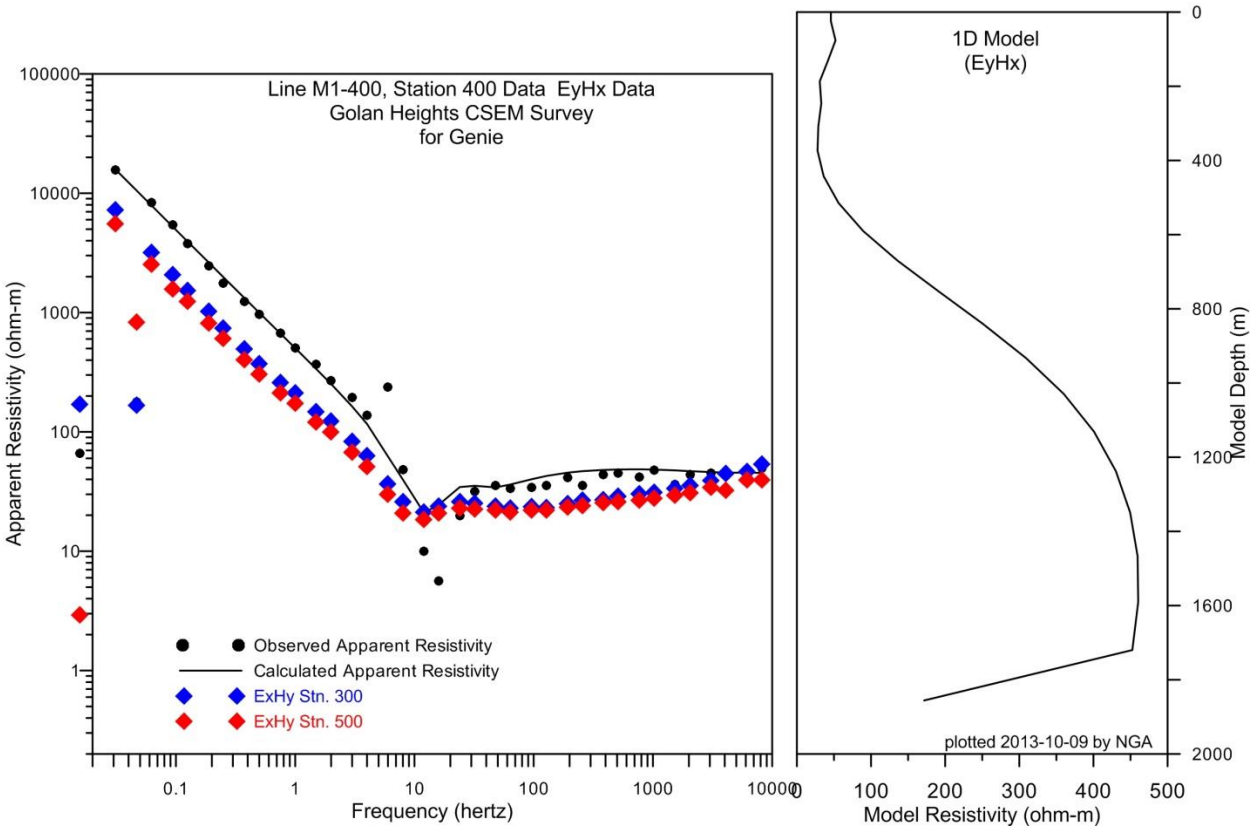


Figure 17: Orthogonal measurements of resistivity along line M1 support a 1D interpretation.

CONCLUSIONS

The CSAMT survey on the Afek Project produced good results despite terrain and cultural obstacles. The interpretation of the CSAMT data appears consistent with previous geophysical investigations. The resistivity lines were short enough to permit 1D CSAMT modeling, but when examined collectively, they tell a coherent 3D story. The results indicate that the target interval has higher resistivities in the eastern part of the survey area, likely indicating more thermally mature sections, and that toward the west the target zone is either less resistive (less thermally developed) or is too deep to reliably resolve.

DATA DELIVERABLES

A report archive disk includes the following:

- Report document (PDF). Note the report Appendices are in a separate PDF file.
- Survey location maps showing transmitter, line, and station locations
- Scalar-mode CSAMT plots
  - 1D inversion model resistivity profile sections
  - 4-panel plots with Observed and Calculated pseudosections
  - Cagniard Resistivity and Impedance Phase pseudosections
- 3D fence plots of resistivity profiles
- Raw, processed and modeled data
- Equipment specifications

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